



EAST WEST UNIVERSITY

Lab Report-2

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Section: 01

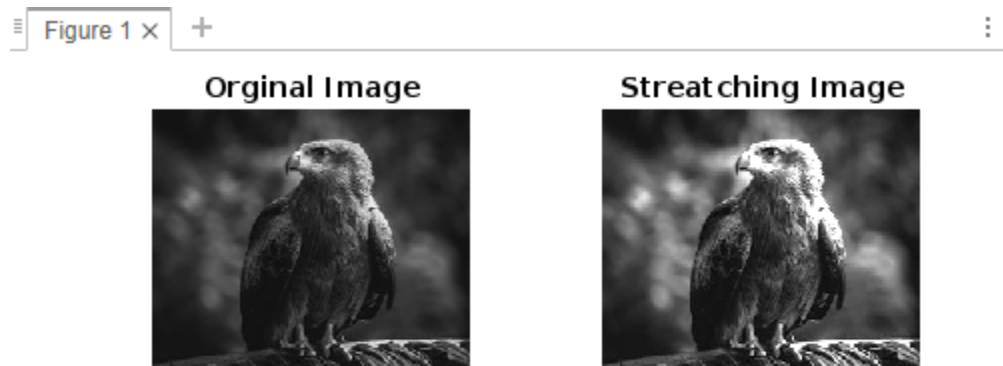
1. Use contrast stretching on the image.



Answer:

```
i = imread('C:\Users\user\Pictures\Picture1.jpg');  
hold = imadjust(i, [0.03, 0.60], [0,1]);  
imshow(hold)  
subplot(2,2,1), imshow(i), title('Original Image');  
subplot(2,2,2), imshow(hold), title('Strectching Image');
```

Output:



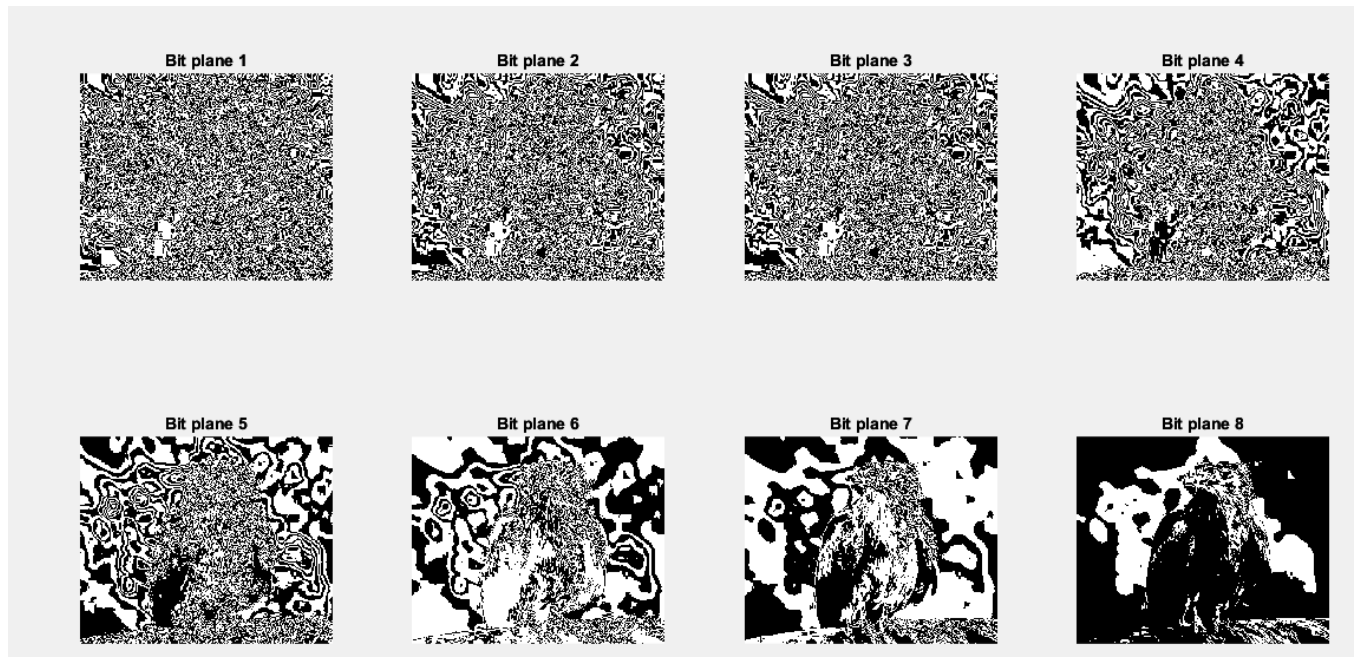
2. Apply bit plane slicing on the image.



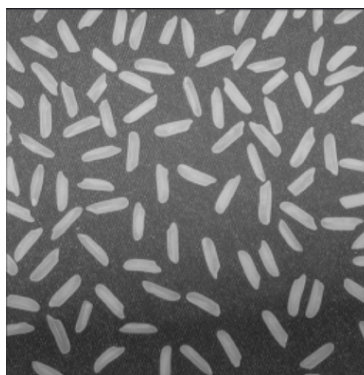
Answer:

```
img = imread('C:\Users\user\Pictures\Picture7.jpg');  
img= double(img);  
j= bitget(img,1);  
subplot(2,4,1), imshow(j); title('Bit plane 1');  
j=bitget(img,2);  
subplot(2,4,2), imshow(j); title('Bit plane 2');  
j=bitget(img,2);  
subplot(2,4,3), imshow(j); title('Bit plane 3');  
j=bitget(img,3);  
subplot(2,4,4), imshow(j); title('Bit plane 4');  
j=bitget(img,4);  
subplot(2,4,5), imshow(j); title('Bit plane 5');  
j=bitget(img,5);  
subplot(2,4,6), imshow(j); title('Bit plane 6');  
j=bitget(img,6);  
subplot(2,4,7), imshow(j); title('Bit plane 7');  
j=bitget(img,7);  
subplot(2,4,8), imshow(j); title('Bit plane 8');  
j=bitget(img,8);
```

Output:



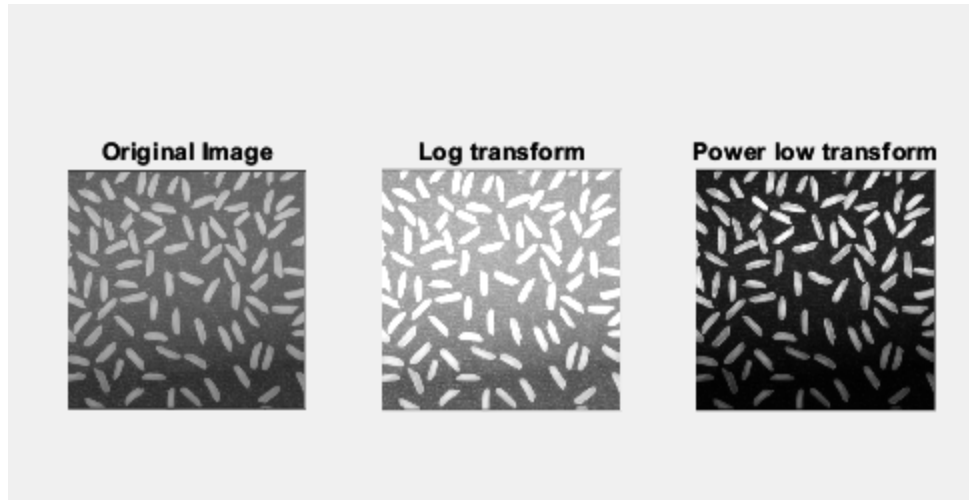
3. Change the contrast of the image using Logarithmic Transformation and Power-law Transformation.



Answer:

```
img = imread('C:\Users\user\Pictures\Picture3.png');
gray_img= rgb2gray(img);
double= im2double(gray_img);
output1= 2*log(1 + double);
subplot(1,3,1), imshow(gray_img); title('Original Image');
subplot(1,3,2), imshow(output1); title('Log transform');
output2= 2*(double.^ 3);
subplot(1,3,3), imshow(output2); title('Power low transform');
```

Output:



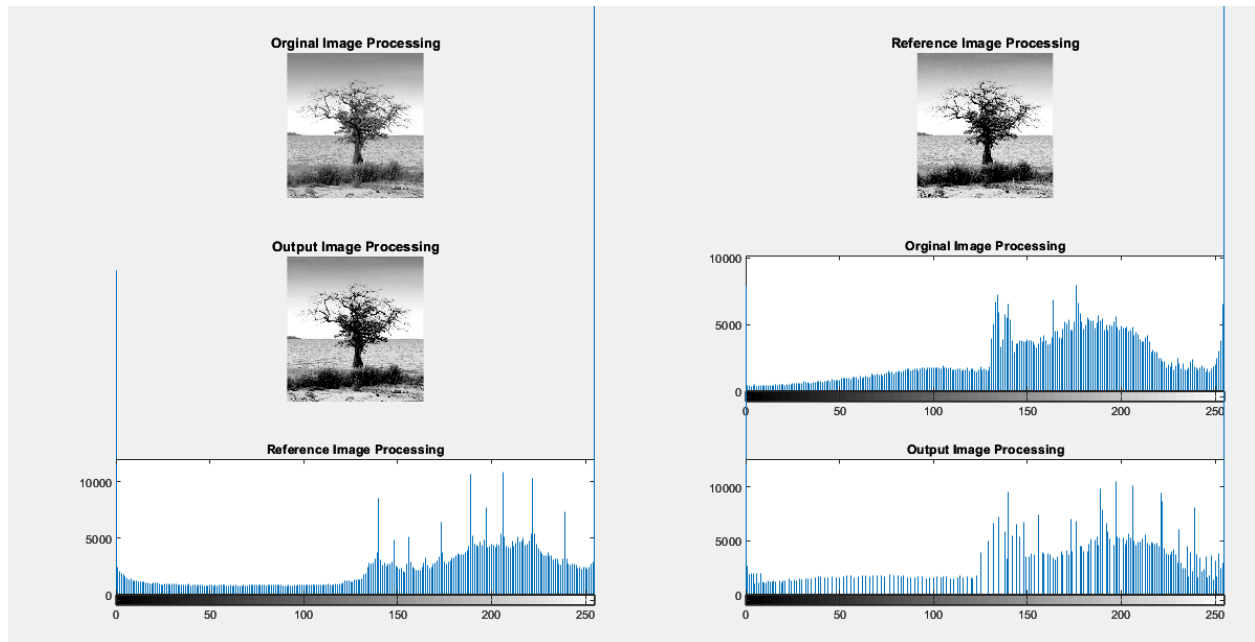
4. Adjust the histogram of the following image to match the reference image using histogram matching. Show the histogram of original, reference, and output images.



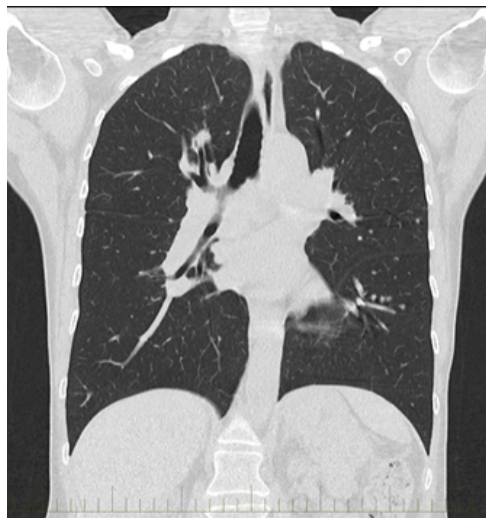
Answer:

```
img= imread('C:\Users\user\Pictures\Picture4.jpg');
reference_img = imread('C:\Users\user\Pictures\Picture5.png');
output_img = histeq(img, imhist(reference_img));
subplot(3,2,1), imshow(img), title('Original Image Processing');
subplot(3,2,2), imshow(reference_img), title('Reference Image Processing');
subplot(3,2,3), imshow(output_img), title('Output Image Processing');
subplot(3,2,4), imhist(img), title('Original Image Processing');
subplot(3,2,5), imhist(reference_img), title('Reference Image Processing');
subplot(3,2,6), imhist(output_img), title('Output Image Processing');
```

Output:



5. Change the contrast of the image using histogram equalization. Show the histogram of both input and output images.



Answer:

```
img= imread('C:\Users\user\Pictures\Picture6.jpg');
output_img = histeq(img);
subplot(2,1,1), imhist(img), title('Input Histogram');
subplot(2,1,2), imhist(output_img), title('Output Histogram');
figure;
subplot(1,2,1), imshow(img), title('Input Image');
subplot(1,2,2), imshow(output_img), title('Output Image');
```

Output:

